
**Petroleum products — Determination of
sulfur content — Energy-dispersive X-ray
fluorescence spectrometry**

*Produits pétroliers — Détermination de la teneur en soufre —
Spectrométrie de fluorescence de rayons X dispersive en énergie*



PDF disclaimer

This PDF file may contain embedded typefaces. In accordance with Adobe's licensing policy, this file may be printed or viewed but shall not be edited unless the typefaces which are embedded are licensed to and installed on the computer performing the editing. In downloading this file, parties accept therein the responsibility of not infringing Adobe's licensing policy. The ISO Central Secretariat accepts no liability in this area.

Adobe is a trademark of Adobe Systems Incorporated.

Details of the software products used to create this PDF file can be found in the General Info relative to the file; the PDF-creation parameters were optimized for printing. Every care has been taken to ensure that the file is suitable for use by ISO member bodies. In the unlikely event that a problem relating to it is found, please inform the Central Secretariat at the address given below.

© ISO 2003

All rights reserved. Unless otherwise specified, no part of this publication may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm, without permission in writing from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
Case postale 56 • CH-1211 Geneva 20
Tel. + 41 22 749 01 11
Fax + 41 22 749 09 47
E-mail copyright@iso.org
Web www.iso.org

Published in Switzerland

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

International Standards are drafted in accordance with the rules given in the ISO/IEC Directives, Part 2.

The main task of technical committees is to prepare International Standards. Draft International Standards adopted by the technical committees are circulated to the member bodies for voting. Publication as an International Standard requires approval by at least 75 % of the member bodies casting a vote.

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. ISO shall not be held responsible for identifying any or all such patent rights.

ISO 8754 was prepared by Technical Committee ISO/TC 28, *Petroleum products and lubricants*.

This second edition cancels and replaces the first edition (ISO 8754:1992), which has been technically revised.

Introduction

Specialized procedures, using the analytical technique described in this International Standard, for automotive fuels with sulfur contents below 0,20 % (*m/m*), are under development.

Petroleum products — Determination of sulfur content — Energy-dispersive X-ray fluorescence spectrometry

WARNING — The use of this International Standard may involve hazardous materials, operations and equipment. This International Standard does not purport to address all of the safety problems associated with its use. It is the responsibility of the user of this International Standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

1 Scope

This International Standard specifies a method for the determination of the sulfur content of petroleum products, such as naphthas, unleaded motor gasolines, middle distillates, residual fuel oils, base lubricating oils and components. The method is applicable to products having sulfur contents in the range 0,03 % (m/m) to 5,00 % (m/m).

NOTE For the purposes of this International Standard, the term "% (m/m)" is used to represent the mass fraction of a material.

Heavy metal additives, such as lead alkyls, may interfere with the determination. Elements such as silicon, phosphorus, calcium, oxygen, potassium, zinc, molybdenum, barium and halogens interfere, if present in concentrations of more than a few hundred milligrams per kilogram. Some modern instruments allow the analyst to compensate for matrix and spectral interferences by spectra deconvolution and inter-element correction by multiple regression.

For samples varying in composition of aromatic hydrocarbons and paraffinic hydrocarbons, the ratio of carbon to hydrocarbon in a sample (C/H ratio) may also interfere with the determination, when the ratio of the sample differs by one or more from that of the reference materials from which the calibration is obtained.

2 Normative references

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 3170:—¹⁾, *Petroleum liquids — Manual sampling*

ISO 3171:1988, *Petroleum liquids — Automatic pipeline sampling*

3 Principle

The test portion is placed in the beam emitted from an X-ray source. The excitation energy may be derived from a radioactive source, such as ⁵⁵Fe, or from an X-ray tube. The resultant excited characteristic X radiation is measured, and the accumulated count is compared with a calibration graph plotting counts against sulfur content as percentage by mass [% (m/m)], on a series of calibration samples covering the range of sulfur contents under examination.

1) To be published. (Revision of ISO 3170:1988)

4 Reagents and materials

4.1 White oil (light paraffin oil), of high purity grade, with a maximum sulfur content of 20 mg/kg.

For measurements at very low levels of sulfur content [below approximately 0,1 % (m/m)], matrix effects are minimized if a diluent of a type similar to the product being analysed is used for the production of the calibration standards described in Clause 8. Such a diluent should be of very low sulfur content, preferably below 2 mg/kg.

4.2 Sulfur compounds, of known sulfur content, used for the preparation of the primary standards.

NOTE The compounds given in 4.2.1 to 4.2.3 are suitable, and their nominal sulfur contents are given. Where the purity of these compounds is less than 99 %, certified materials are required, or the concentrations and nature of all impurities are to be known.

4.2.1 Dibenzothiophene (DBT), with a nominal sulfur content of 17,399 % (m/m).

4.2.2 Dibutylsulfide (DBS), with a nominal sulfur content of 21,915 % (m/m).

4.2.3 Thionaphthene (benzothiophene) (TNA), with a nominal sulfur content of 23,89 % (m/m).

4.3 Certified reference materials: use materials from a national standards body or accredited supplier, with a range of certified sulfur contents for the production of calibration curves for routine analysis. However, particularly for some heavier materials, the excitation is slightly affected by the background matrix. Therefore, in cases of dispute, the parties shall agree on a common range of certified standards, or shall both prepare standards from the materials described in 4.1 and 4.2.

5 Apparatus

5.1 Energy-dispersive X-ray fluorescence analyser: use any suitable model, provided that the design incorporates the features given in 5.1.1 to 5.1.6. It shall be set up according to the manufacturer's instructions.

5.1.1 Source of X-ray excitation, with significant X-ray flux at energies above 2,5 keV.

5.1.2 Removable sample cup, providing a sample depth of at least 3 mm, and equipped with replaceable X-ray transparent film.

NOTE Window material is normally 6 µm polyester, polypropylene or polycarbonate film. Commercial polyester film may contain small but variable amounts of calcium, which may interfere. Samples of very high aromatic content may dissolve polycarbonate film.

5.1.3 X-ray detector, with high sensitivity at 2,3 keV.

5.1.4 Filters, or other means of discriminating between sulfur K α radiation and other X-rays.

5.1.5 Signal-conditioning electronics, that include the functions of pulse counting and pulse-height analysis.

5.1.6 Display or printer, that provides a readout in counts, sulfur content as a percentage by mass [% (m/m)], or both.

CAUTION — If the analyser contains a radioactive source, the equipment and manner of use shall comply with the regulations governing the use of ionizing radiation and/or recommendations of the International Commission on Radiological Protection. The radiation source shall be checked for radiation leakage at intervals as required by the regulations. All attention to the source shall only be carried out by fully trained and competent persons, using the correct shielding techniques.

5.2 Analytical balance, capable of weighing to the nearest 0,1 mg.

5.3 Mixer, of non-aerating, high-speed shear type.

5.4 Containers: either flasks of 100 ml capacity, narrow-necked, conical, and made of borosilicate glass, with optional stoppers for use with volatile samples, or bottles of similar capacity fitted with polyethylene or polytetrafluoroethylene seals in the closure.

6 Samples and sampling

6.1 Unless otherwise specified, samples shall be taken in accordance with the procedures described in ISO 3170 or ISO 3171.

6.2 Test portions from the samples shall be drawn after thorough mixing and subdivision. Heat viscous samples to a temperature which renders the sample liquid, and homogenize, using the mixer (5.3) as necessary.

7 Apparatus preparation

7.1 Analyser

Wherever possible, the instrument (see 5.1) shall be continuously switched on to maintain optimum stability.

7.2 Sample cups

Thoroughly clean and dry the sample cups before use. Do not re-use disposable cups. Keep handling of window material to the absolute minimum.

8 Calibration

8.1 General

Use either certified reference materials (4.3) or primary standards prepared from sulfur compounds (4.2) dissolved in white oil (4.1) as a basis for the preparation of the appropriate range of calibration standards.

8.2 Preparation of primary standards

8.2.1 Prepare a range of at least three primary standards. Either use primary standards to prepare the calibration standards by dilution with white oil (4.1), or prepare the complete range of calibration standards directly by the procedure given in 8.2.2.

8.2.2 Weigh, to the nearest 0,1 mg, the appropriate quantity of white oil (see Table 1), into a container (5.4) and add the appropriate quantity of the selected sulfur compound (4.2), weighed to the nearest 0,1 mg. Mix the contents of the container thoroughly at room temperature.

It is recommended that an inert magnetic stirrer and stirring device are used to mix the contents of the container. In the case of volatile liquids, the container should be stoppered, and the mixture gently shaken or agitated.

Calculate the exact sulfur content, w_S , expressed as a percentage by mass [% (m/m)] to three decimal places in each case, from the amounts of white oil and sulfur compound used as follows:

$$w_S = \frac{m_1 \times w_{S,c}}{m_1 + m_2}$$

where

m_1 is the mass of sulfur compound, expressed in grams;

$w_{S,c}$ is the sulfur content of the sulfur compound, expressed as a percentage by mass;

m_2 is the mass of white oil, expressed in grams.

Table 1 — Composition of primary standards based on nominal sulfur contents

Approximate sulfur content % (m/m)	White oil g	DBT (4.2.1) g	DBS (4.2.2) g	TNA (4.2.3) g
5	40,0	16,1	11,8	10,6
4	40,0	11,95	8,95	8,05
3	40,0	8,3	6,3	5,75
2	45,0	5,85	4,5	4,1
1,5	45,0	4,25	3,3	3,0
1	45,0	2,75	2,15	1,95
0,5	50,0	1,5	1,15	1,05
0,3	50,0	0,9	0,7	0,65
0,1	50,0	0,3	0,25	0,2
0,07	50,0	0,21	0,18	0,14
0,05	50,0	0,14	0,11	0,1
0,03	50,0	0,09	0,07	0,07

8.3 Calibration standards

For analysers without curvature fitting, use calibration standards in four ranges (see Table 2). For analysers with curvature fitting, or those which require manual graph drawing, combine two adjacent ranges, using all the standards within those ranges.

Table 2 — Calibration standard ranges

Range	Sample sulfur content % (m/m)	Standard sulfur content % (m/m)				
1	0,03 to 0,10	0,00	0,03	0,05	0,07	0,10
2	> 0,10 and ≤ 0,5	0,00	0,1	0,3	0,5	
3	> 0,5 and ≤ 2,0	0,5	1,0	1,5	2,0	
4	> 2,0 and ≤ 5,0	2,0	3,0	4,0	5,0	

8.4 Storage of standards

Store certified reference standards in accordance with the instructions of the certifying organization, and use within the timescale specified.

Store primary and calibration standards, prepared from white oil and sulfur compounds, in dark glass-stoppered bottles in a cool dark place.

NOTE The stability of stored standards, under the above storage conditions, has been found to exceed three months.

8.5 Calibration procedure

Prepare the sample cup (5.1.2) in accordance with the manufacturer's instructions. Cover the base of the cup with window film, avoiding skin contact, and fill to a depth of between 3 mm and 20 mm with the calibration standard. Ensure that the window is smooth and taut and that there are no air bubbles between the window and the liquid.

Obtain a reading on each standard, using the recommended counting time for the instrument.

Usually, it is sufficient to obtain a single reading for each standard. However, it may be advisable to immediately repeat the measuring procedure up to four times, using freshly prepared cups and fresh portions of the standard, if there is an indication of erratic or erroneous results.

Taking consecutive single readings using the same cup and standard portion is not recommended. Cups and films will change shape as they are measured, or waiting to be measured. In addition, samples may be volatile and change composition with time.

From the data obtained, calculate the average, if necessary, for each sulfur content. From the resulting data set, construct the calibration curve by linear regression, either manually or with the aid of a computer, using sulfur content as an independent variable and the readings as dependent variables.

8.6 Checking

8.6.1 Short-term checking

In regular use, check at least two points on each calibration curve not less frequently than weekly. This shall be accomplished by using set-up standards, which have been assigned and selected from an inspection of the calibration curve. If the check results differ from the curve by more than the repeatability, repeat the calibration procedure.

8.6.2 Long-term checking

Carry out more elaborate checks at intervals not greater than three months, following the procedure described in 8.5.

9 Procedure

Prepare and fill the sample cup with the test portion as described in 8.5, and obtain two counts using the manufacturer's recommended counting time for the instrument.

For volatile samples, care should be taken to prevent window bowing if the sample cup is capped. Venting may be necessary to avoid excessive pressure build-up, but if losses due to evaporation are noted during measurement, a fresh portion of sample in a new cup should be used for the second count, and the counting time reduced, even at the expense of counting precision.

Calculate the average count for the sample.

NOTE Modern instruments may incorporate this calculation.

10 Calculation

Read the concentration of sulfur in the sample from the calibration curve, using the average count for each test portion, or by direct reading from those analysers which have computing facilities.

11 Expression of results

Report the result to the nearest 0,01 % (m/m) sulfur content for values between 0,10 % (m/m) and 5,00 % (m/m), and to the nearest 0,001 % (m/m) sulfur content for values between 0,030 % (m/m) and 0,099 % (m/m).

12 Precision

12.1 General

The precision was determined by statistical examination, in accordance with ISO 4259:1992, *Petroleum products — Determination and application of precision data in relation to methods of test*, of interlaboratory test results on a matrix of samples including gasolines, kerosines, gas oils and residual fuel oils with sulfur contents from below 0,10 % (m/m) to above 5,0 % (m/m).

NOTE The precision given in 12.2 and 12.3 was developed by analysis of the combined test results from three large precision-evaluation programmes, one sponsored by ASTM in the USA in 1995/1996 and one sponsored by the IP in Europe in 1997 on the range of fuels covering the whole scope of the test method, and the third sponsored by AFNOR in France covering automotive gasolines and diesel fuels of low sulfur content only. Both instrument types were included in all three programmes.

12.2 Repeatability

The difference between two test results obtained by the same operator with the same apparatus under constant operating conditions on identical test material would, in the long run, in the normal and correct operation of the test method, exceed the following values in only one case in twenty:

$$r = 0,045\,4 (X + 0,05), \text{ for values } \geq 0,03 \% (m/m) \text{ and } \leq 0,05 \% (m/m);$$

$$r = 0,021\,5 (X + 0,15), \text{ for values } > 0,05 \% (m/m) \text{ and } \leq 5,00 \% (m/m);$$

where X is the average of the results being compared, in % (m/m).

12.3 Reproducibility

The difference between two single and independent test results obtained by different operators working in different laboratories on identical test material would in the long run, in the normal and correct operation of the test method, exceed the following values in only one case in twenty:

$$R = 0,178\,1 (X + 0,05), \text{ for values } \geq 0,03 \% (m/m) \text{ and } \leq 0,05 \% (m/m);$$

$$R = 0,081\,2 (X + 0,15), \text{ for values } > 0,05 \% (m/m) \text{ and } \leq 5,00 \% (m/m);$$

where X is the average of the results being compared, in % (m/m).

13 Test report

The test report shall contain at least the following information:

- a) a reference to this International Standard;
- b) the type and complete identification of the product tested;
- c) the result of the test (see Clause 11);
- d) any deviation, by agreement or otherwise, from the procedure specified;
- e) the date of the test.

